SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED
CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2) Total Marks: 25
Question Count: 5

Code of Conduct
I G. Amrutha (192511250) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Amutha.

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

2. An organization wants to store large volumes of backup data securely in the cloud.
   Discuss how cloud storage solutions ensure durability and data redundancy.
   How can encryption and access control protect sensitive backup data?
   Explain the cost advantages compared to traditional storage systems.
   Provide an example of a disaster recovery use case.

1) Auto-Scaling and Load Balancing

Auto-Scaling: Automatically increases or decreases servers based on traffic.

Load balancing: Distributes requests across multiple servers

Benefits: Reduces downtime, improves speed, and saves costs.

Monitoring: Tracks CPU, memory, and errors

Ex: Amazon handles festival sale traffic using scalable cloud infrastructure.

2) Secure Cloud Backup Storage

Durability: Stores data reliably to prevent loss

Redundancy: maintains multiple copies of data

Security: Encryption and access controls protect sensitive backups.

Cost: Reduces hardware and maintenance expenses.

Example: A company restores cloud backups after a server failure.

3. Content Delivery Network (CDN)

=) A CDN stores content on servers near users.

=) Edge servers reduce latency and buffering

=) Cloud regions improve global availability.

=) CDNs reduce the load on origin servers.

Ex: Netflix uses content delivery infrastructure for smooth video streaming.

4. Cloud security and Compliance

=) Cloud providers offer compliance tools and audit reports.

=) Logging records user activities; monitoring detects threats.

=) Encryption protects data during storage and transmission.

=) Data sovereignty is maintained by selecting approved storage regions

Ex: A bank encrypts transactions and monitors suspicious activities

5. Containers and kubernetes

Containers: Lightweight, fast, and share the host OS kernel.

Virtual machines: Run separate guest operating systems and generally use more resources.

Kubernetes: Automates deployment, scaling and recovery.

Portability:

Containers can run across compatible cloud environments.

Ex:

An e-commerce application uses separate microservices for payments, orders and user login.

3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity